Current: charge flow rate

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate current as charge transferred per time.

SUCCESS CRITERIA - I can...

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Both charge magnitudes become 3 times as large and separation becomes 5 times as large. Express new force as a fraction of the old.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: I = Δq/Δt.
Current is charge per time; 1 A = 1 C/s. Conventional current follows positive charge motion.

Worked reasoning: If 12 C pass a point in 4 s, average current is I=Q/t=3 A. Charge circulates; a resistor does not consume charge.

Your turn: Draw a complete loop and label equal current before and after one resistor in steady state.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw conventional current around the closed circuit, outside the battery from + toward -. What happens if a wire is disconnected?



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

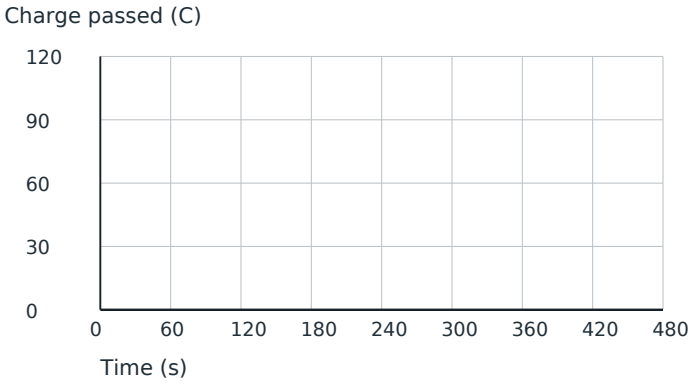
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Continue down each column.

04 Apply the physics

A wire carries a steady 0.25 A for 8 minutes. Convert time to seconds, then calculate charge transferred. Draw a cumulative-charge graph and label the slope.



Marzano check - circle: 1 2 3

05 Explain your reasoning

At a junction, 5 A enters and 2 A leaves along one branch. Find the other outgoing current in steady operation. Explain the conservation rule.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A source approaching a stationary listener at 20 m/s emits 600 Hz. Use $v_{sound}=340$ and $f_{obs}=f_s\ v/(v-v_s)$. Find observed frequency.
Use wavefront spacing to explain the frequency change.

Source moves right. Add wavefronts.



Marzano check - circle: 1 2 3

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Continue down each column.

07 Exit check

Does a large electron drift speed have to equal the speed at which a switched circuit responds? Explain qualitatively.

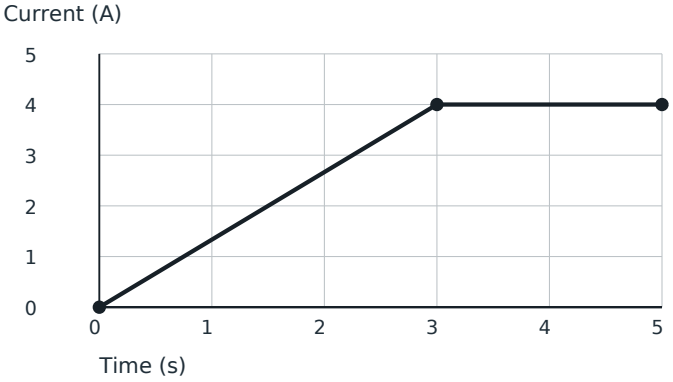


Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A current rises linearly from 0 to 4 A during 3 s, then stays 4 A for 2 s. Find total charge from the current-time graph.



Marzano check - circle: 1 2 3